Soroush Azari

Data Analyst (Power BI, SQL, Python)

Portfolio: azed.uk · **Code:** github.com/su7ri

Exeter, UK · willing to relocate

Profile

Data analyst with an MSc in Advanced Computer Science (Merit) and hands-on experience taking large datasets from several sources through cleaning, quality checks and analysis to Power BI dashboards and written reports. My most recent project combines Environment Agency rainfall and river-flow records with South West Water's storm-overflow monitoring data: about 36,000 daily readings and 209,711 overflow events, with a next-day flow forecast, anomaly checks and limitations stated plainly. I like explaining findings to non-technical readers, and I want to apply this to water-resources reporting. Projects and code: azed.uk.

Projects

Rainfall, River Flow and Storm Overflows in Devon and Cornwall (Python, SQL, Power BI) · azed.uk/water.html

- Brought together two external sources: 2015–2025 daily rainfall (7 gauges) and river flow from the Environment Agency Hydrology API, and South West Water's 2025 storm-overflow monitoring data (209,711 events at 1,179 sites).
- Ran quality assurance on the data: resolved duplicate readings, flagged the 10.7% of readings not rated Good, and excluded 356 implausibly long overflow events, documenting each decision.
- Analysed trends, patterns and anomalies: the last 14 days of rain explain River Exe flow far better than same-day rain (correlation 0.77 against 0.34); overflow events were about six times as frequent on wet days; the worst 10% of sites account for about 47% of discharge hours; low-flow days cluster in 2022, 2018 and 2025.
- Built a next-day flow forecast (Ridge regression) that beats a persistence baseline on held-out data (Nash-Sutcliffe efficiency 0.85 against 0.78).
- Built a six-page Power BI dashboard with DAX measures, backed by 11 SQL queries on a star schema, and wrote a short report with the limitations stated plainly. The notebook, SQL and report are public.

Credit Risk Analytics and Policy Scenario Analysis (SQL, Python, Power BI) · azed.uk/credit-risk.html

- Built a customer-level dataset by joining several relational tables and engineered affordability and behavioural features for risk segmentation.
- Analysed portfolio and segment risk, finding default risk concentrated in particular customer groups.
- Modelled approval-versus-risk trade-offs under different credit thresholds and built an interactive dashboard of risk exposure and policy outcomes.

UK Railway Performance Analysis (Power BI, SQL, Python) · azed.uk/railway.html

- End-to-end analysis of rail performance and ticketing data, with dashboards tracking delays, pricing and route performance.

Core technical skills

- **Reporting and BI:** Power BI (data modelling, DAX measures, dashboards), Excel
- **Data analysis:** SQL (joins, window functions, CTEs), Python (pandas, scikit-learn, regression and forecasting)
- **Practice:** data cleaning and quality assurance, trend and anomaly analysis, KPI definition, writing up findings and limitations for non-technical readers
- **Open data:** Environment Agency Hydrology API, open government datasets
- **Tools:** Git and GitHub, basic cloud exposure

Education

- **MSc Advanced Computer Science**, University of Exeter (Merit), 2023–2024. Dissertation: balancing accuracy and fairness in logistic regression. Modules: Machine Learning; Learning from Data; Social Networks and Text Analysis; Computer Vision; High-Performance Computing; Research Methodology.
- **MSc Computer Engineering (Artificial Intelligence and Robotics)**, Iran, 2015–2018. Modules: Machine Learning; Pattern Recognition; Neural Networks; Advanced Data Mining.
- **BSc Computer Engineering (Software)**, Islamic Azad University, Lahijan, 2010–2015. Included probability and statistics, mathematics and databases.